MA378 Chapter 3: Numerical Integration

§3.5 Orthogonal Polynomials

Dr Niall Madden
(Niall.Madden@UniversityOfGalway.ie)

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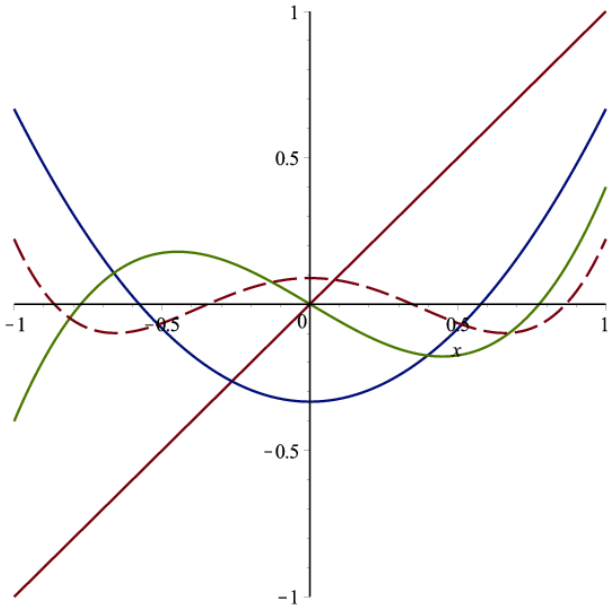

5.0 The span of topics for today:

1. Orthogonal Polynomials
2. Inner products
   - Properties
3. Sequence of monic polynomials
4. Constructing the Sequence
5. Properties of the sequence
6. Legendre Polynomials
7. Exercises

5.1 Orthogonal Polynomials

High order Newton-Cotes methods are of little use because of the problems associated with interpolation by high degree polynomials at equally spaced points. However, high-order Gaussian methods are very useful.

Deriving such methods by undetermined coefficients is not practical, however. There is a simpler way, but some mathematical preliminaries are required, including the ideas of **vector spaces** and **inner products**.

Definition 5.1 (Vector space: informal)

A **vector** space is a collection of objects, called **vectors**, where it makes sense to

- add two vectors to get another one;
- multiply a vector by a scalar to get another one.

5.2 Inner products

Definition 5.2 (Vector Space: formal)

V is a *vector space* (a.k.a., a *linear space*) over a field F (e.g, the real or complex numbers) if for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $a, b \in F$:

(i) $\mathbf{u} + \mathbf{v} \in V$ (closed under addition)

(ii) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (Commutativity)

(iii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (Associativity)

(iv) V has a zero vector $\mathbf{0}$ such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$.

(v) $-\mathbf{u} \in V$

(vi) $a\mathbf{u} \in V$

(vii) $a(b\mathbf{u}) = (ab)\mathbf{u}$

(viii) F contains 0 and 1 such that $1\mathbf{u} = \mathbf{u}$, $0\mathbf{u} = \mathbf{0}$.

(ix) $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$, and $(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$.

5.2 Inner products

Examples:

1. Vectors in $\mathbb{R}^3$

$$u = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \qquad v = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \qquad w = \begin{pmatrix} -\pi \\ e \\ 12 \end{pmatrix}$$

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2. The vector space, $\mathbb{P}_n$, of polynomials of degree n.

Eg P_2.

$u = x^2 \qquad v = 3x^2 - 5.1 \qquad w = x^2 - x - 1$

$z = 3x + 12.5$

For us, here are the most important facts about (real) vector spaces:

Linear combination: Given any set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m\} \in V$ we call $\mathbf{w} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_m\mathbf{v}_m$ a **linear combination**. Note: $\mathbf{w} \in V$.

Spanning Set: The set of all possible linear combinations of the vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_m$ is their **spanning set**.

Linear Independence: A set of vectors is **linearly independent** in no element is a linear combination of the others.

Basis: If the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ spans all of V, and its linearly independent, then it is called a **Basis** of V.

Dimension: **FACT:** a vector space, V, will have many bases, but they all have the same number of elements. That number is called the **dimension** of V.

Definition 5.3 (Inner Product)

Let V be a real vector space. An **Inner Product** (IP) is a real-valued function $(\cdot,\cdot)$ on $V \times V$ such that, for all $f, g, h \in V$,

(i) $(f+g,h) = (f,h) + (g,h)$,
(ii) $(\lambda f, g) = \lambda(f,g)$, for $\lambda \in \mathbb{R}$.

Linearity

(iii) $(f,g) = (g,f)$, → Symmetry

(iv) $(f,f) \geq 0$. $(f,f) = 0 \Leftrightarrow f \equiv 0$. → "positivity".

Example 5.4

Let $\mathbb{R}^n$ be our vector space, with $\mathbf{x} = (x_1, x_2, \ldots, x_n)^T$ and $\mathbf{y} = (y_1, y_2, \ldots, y_n)^T$. Then the following is an inner product:

$$(\mathbf{x}, \mathbf{y}) := \sum_{i=1}^{n} x_i y_i,$$

Eg $n=2$.

$x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ $(x,x) = x_1^2 + x_2^2 = 2$.

$y = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ $(x,y) = x_1 y_1 + x_2 y_2 = 0$

Example 5.5

The set of real-valued functions that are continuous and defined on the interval $[a, b]$, denoted $C[a, b]$, is a vector space. And

$$(f, g) := \int_a^b f(x)g(x)dx, \tag{1}$$

is an inner product.

Are we sure this is an IP? Check!

(i) $(f+g, h) = \int_a^b (f+g)\,h\,dx = \int_a^b f\,h\,dx + \int_a^b g\,h\,dx$

$= (f,h) + (g,h)$ ✓

(iv) $(f,f) = \int_a^b (f(x))^2\,dx \geq 0.$ And $(f,f) = 0 \iff f \equiv 0$

5.3 Sequence of monic polynomials

(See Lecture 23 of Stewart's "Afternotes" for more details).

Definition 5.6 (Monic Polynomial)

A polynomial is *monic* if the coefficient of its leading term is 1.

Examples: $p(x) = 1$ is monic

$p(x) = x + 2$, $q(x) = 3x^2 - x^5 + x^9 + 34$

are monic. So are $\{1, x, x^2, x^3, \ldots\}$

Not monic:

$p(x) \equiv 2$

$q(x) = x + 2 + 3x^2$

$r(x) = 3x^2 - x^5 - x^9 + 34$

5.3 Sequence of monic polynomials

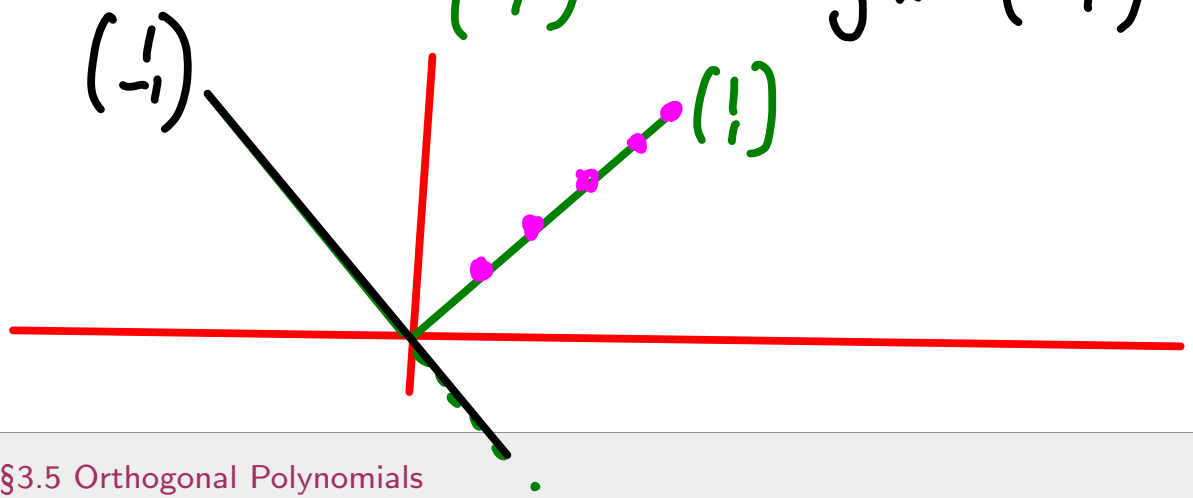
Definition 5.7

Two elements a, b, of a vector space are *orthogonal* with respect to a given inner product $(\cdot, \cdot)$ if $(a, b) = 0$.

Example: In $\mathbb{R}^3$ $\quad u = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} \quad v = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$

These are orthogonal $\quad (u, v) = (1)(-2) + (2)(1) = 0.$

In $\mathbb{R}^2$: $\quad x = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad y = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad (x, y) = 0$



Example 5.8

Take the space of polynomials of degree 2 or less and the IP

$$(f, g) = \int_{-1}^{1} f(x)g(x)dx.$$

Let $p(x) \equiv 1$, $q(x) \equiv x$, $r(x) \equiv x^2 - 1/3$, and $f(x) = 3x - 4$

We can check that $(r, p) = 0$, and $(r, q) = 0$. We can then verify that $(r, f) = 0$. **Details:**

$$(r,p) = \int_{-1}^{1} (x^2 - \tfrac{1}{3})\,dx = \left[\tfrac{1}{3}x^3 - \tfrac{1}{3}x\right]_{-1}^{1} = \tfrac{1}{3} - \tfrac{1}{3} + \tfrac{1}{3} - \tfrac{1}{3} = 0$$

$$(r,q) = \int_{-1}^{1} x^3 - \tfrac{1}{3}x\,dx = \tfrac{1}{4}x^4 - \tfrac{1}{6}x^2 \Big|_{-1}^{1} = \tfrac{1}{4} - \tfrac{1}{4} - \tfrac{1}{6} + \tfrac{1}{6} = 0.$$

Also $f(x) = 3q(x) - 4p(x)$. So $(r,f) = 3(r,q) - 4(r,p) = 0$

5.3 Sequence of monic polynomials

As given above, a polynomial is **monic** if the coefficient of the leading term is 1:

$$p_n = x^n + c_{n-1}x^{n-1} + c_{n-2}x^{n-2} + \cdots + c_1 x + c_0.$$

We'll now look at a sequence of such polynomials

$$\{\widetilde{p}_0, \widetilde{p}_1, \widetilde{p}_2, \ldots, \widetilde{p}_n, \ldots\}$$

that are orthogonal to each other:

$$(\widetilde{p}_i, \widetilde{p}_j) := \int_a^b \widetilde{p}_i(x)\widetilde{p}_j(x)dx = 0 \quad \text{if } i \neq j.$$

5.3 Sequence of monic polynomials

We want to establish some important facts about monic polys:

- A set of monic polynomials of degrees $1, 2, \ldots, n$, forms a basis for $\mathcal{P}_n$.
- If the members of that set are orthogonal to each other, then they are orthogonal to *all* polynomials of lower degree.
- We can construct such as set.

5.3 Sequence of monic polynomials

Theorem 5.9

Let $\{\widetilde{p}_i\}_{i=0}^{n}$ be a sequence of polynomials where each $\widetilde{p}_i$ is monic and exactly of degree i. This sequence forms a basis for $\mathcal{P}_n$.

Proof: Recall that P_n is the space of all polynomials of degree n or less. Also $\tilde{p}_0(x) \equiv 1$. If $n=0$, then any element of P_0 is just a constant - and so a multiple of $\tilde{p}_0$. So $\tilde{p}_0$ is a basis for P_0.

5.3 Sequence of monic polynomials

See board.

5.3 Sequence of monic polynomials

Theorem 5.9 means that if q is a polynomial of degree n then it can be written uniquely as a linear combination of the $\widetilde{p}_i$:

$$q(x) = \sum_{i=0}^{n} a_i \widetilde{p}_i(x),$$

for some unique choice of the real coefficients a_i.

5.3 Sequence of monic polynomials

Definition 5.10 (Legendre Polynomials)

The sequence $\{\widetilde{p}_i\}_{i=0}^n$ is a sequence of *monic*, *orthogonal* polynomials if each $\widetilde{p}_i$ is monic and *exactly* of degree i and

$$(\widetilde{p}_i, \widetilde{p}_j) = 0 \quad \text{if } i \neq j.$$

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Theorem 5.11

If $\widetilde{p}_j \in \{\tilde{p}_i\}_{i=0}^{\infty}$ then $\widetilde{p}_j$ is orthogonal to all polynomials of degree less than j.

Proof: Let q be any polynomial of degree $j-1$. (or less).

Since $\{\tilde{p}_0, \tilde{p}_1, \dots, \tilde{p}_{j-1}\}$ is a basis for $\mathbb{P}_{j-1}$, we can write

$$q(x) = c_0 \tilde{p}_0 + c_1 \tilde{p}_1 + \dots + c_{j-1} \tilde{p}_{j-1}$$

Then $(q, \tilde{p}_j) = c_0(\tilde{p}_0, \tilde{p}_j) + c_1(\tilde{p}_1, \tilde{p}_j) + \dots + c_{j-1}(\tilde{p}_{j-1}, \tilde{p}_j)$

$$= c_0(0) + c_1(0) + \dots + c_{j-1}(0) = 0$$

5.4 Constructing the Sequence

Theorem 5.12

The sequence $\{\tilde{p}_i\}_{i=0}^{\infty}$ exists and can be constructed as follows: Let α and β be defined as

$$\alpha_{n+1} = \frac{(x\widetilde{p}_n, \widetilde{p}_n)}{(\widetilde{p}_n, \widetilde{p}_n)}, \qquad \text{and} \qquad \beta_{n+1} = \frac{(x\widetilde{p}_n, \widetilde{p}_{n-1})}{(\widetilde{p}_{n-1}, \widetilde{p}_{n-1})},$$

then the sequence is given by

$$\widetilde{p}_0(x) \equiv 1, \qquad \widetilde{p}_1(x) = x - \alpha_1$$

and

$$\widetilde{p}_{n+1}(x) = (x - \alpha_{n+1})\widetilde{p}_n(x) - \beta_{n+1}\widetilde{p}_{n-1}(x),$$

for $n \geq 1$.

The proof uses *Gram-Schmidt Orthogonalization*.

5.4 Constructing the Sequence

Idea: we want $\hat{p}_{n+1}$ to be orthogonal to $\hat{p}_0, \hat{p}_1, \ldots,$ and, specifically, $\hat{p}_{n-1}$ & $\hat{p}_n$.

Use $(\hat{p}_{n+1}, \hat{p}_n) = 0$. Therefore

$$\left((x - \alpha_{n+1}) \hat{p}_n + \beta_{n+1} \hat{p}_{n-1}, \hat{p}_n \right) = 0$$

$$\Rightarrow (x \hat{p}_n, \hat{p}_n) - \alpha_{n+1} (\hat{p}_n, \hat{p}_n) + \beta_{n+1} \underbrace{(\hat{p}_{n-1}, \hat{p}_n)}_{0} = 0$$

$$\Rightarrow \alpha_{n+1} = \frac{(x \hat{p}_n, \hat{p}_n)}{(\hat{p}_n, \hat{p}_n)}$$ as required.

Now use that $(\hat{p}_{n+1}, \hat{p}_{n-1}) = 0$ to get β_{n+1}.

Example 5.13

If we use the inner product $(f, g) := \int_{-1}^{1} f(x)g(x)\,dx$ then the first 3 polynomials in the sequence are:

$$\widetilde{p}_0 = 1, \quad \widetilde{p}_1 = x, \quad \text{and} \quad \widetilde{p}_2 = x^2 - 1/3.$$

Check

$$(\tilde{p}_2, \tilde{p}_1) = \int_{-1}^{1} \left(x^2 - \tfrac{1}{3}\right) x \, dx$$

$$= \int_{-1}^{1} x^3 - \tfrac{1}{3} x \, dx = \left(\tfrac{1}{4}x^4 - \tfrac{1}{6}x^2\right)\Big|_{-1}^{1}$$

$$= \tfrac{1}{4} - \tfrac{1}{4} - \tfrac{1}{6} + \tfrac{1}{6} = 0 \quad \checkmark$$

Example 5.14

Find the zeros of $\widetilde{p}_2$ from Example 5.13.

$$\hat{p}_2 = x^2 - \tfrac{1}{3}.$$

So $\hat{p}_2(x_i) = 0 \Rightarrow x_i^2 = \tfrac{1}{3} \Rightarrow x_i = \pm \frac{1}{\sqrt{3}}$

So zeros are

$$x_0 = -\frac{1}{\sqrt{3}} \qquad x_1 = \frac{1}{\sqrt{3}}$$

5.5 Properties of the sequence

One of the ways of constructing the Gaussian Quadrature rule $G_n(\cdot)$ on $n+1$ is to take the quadrature points as the roots of $\widetilde{p}_{n+1}$.

We know (from the fundamental theorem of algebra) a polynomial of degree $n+1$ has exactly $n+1$ roots in $\mathbb{C}$ up to multiplicity.

However, for the roots of $\widetilde{p}_{n+1}$ to be feasible as quadrature points for estimating $\int_a^b f(x)dx$, they would need special properties:

- We need the roots, $x_0, x, \ldots, x_n$ to be distinct.
- All must be real-valued.
- Need each $x_i \in [a,b]$

5.5 Properties of the sequence

Theorem 5.15

Let $\widetilde{p}_i \in \{\widetilde{p}_i\}_{i=0}^{\infty} = \{\widetilde{p}_0, \widetilde{p}_1, \dots\}$ *be the set of monic polynomials that are orthogonal with respect to the inner product* $(u, v) := \int_a^b u(x)v(x)dx$. *Then:*

(i) *The zeros of each* $\widetilde{p}_i \in \{\widetilde{p}_i\}_{i=0}^{\infty}$ *are simple (not repeated).*

(ii) *All the zeros of* $\widetilde{p}_i$ *are real numbers in the interval* $[a, b]$.

(i) Suppose that $\hat{p}_i$ has a repeated root, c.
That is $\hat{p}_i(x) = (x-c)^2 q(x)$ where
$q(x)$ has degree $i-2$. Since $\deg(\hat{p}_i) > \deg(q)$
we should have $(\hat{p}_i, q) = 0$.
But $((x-c)^2 q, q) = \int_a^b (x-c)^2 q(x) q(x)\, dx$

$$= \int_a^b \left[(x-c)\,q(x)\right]^2 dx > 0$$

since $((x-c)(q(x))^2 \geq 0$ for all x.

Contradiction!

(ii) Next we show all the zeros are in $[a,b]$. Suppose $\hat{q}_i$ has k zeros in $[a,b]$ but $k < i$. So we can write

$$\hat{p}_i(x) = q(x)\, r(x) \quad \text{where } q(x)$$

has k zeros in $[a,b]$ and $r(x)$ has none. So $r(x)$ has degree $< i$ and $r(x)$ does not change sign on $[a,b]$.

5.5 Properties of the sequence

So $(\hat{p}_i, q) = 0$. since $\deg(q) = k < i$

But $(\hat{p}_i, q) = \int_a^b q(x)\, r(x)\, q(x)\, dx$

$$= \int_a^b (q(x))^2 r(x)\, dx > 0$$

Since $q(x)^2 r(x)$ does not change sign on $[a,b]$. $\blacksquare$

5.6 Legendre Polynomials

The polynomials we've looked at are an example of **Legendre Polynomials**.

As we'll as been useful for numerical integration methods, they have many other applications, including in astronomy, electrical engineering, number theory, and neural networks.

5.6 Legendre Polynomials

FYI: The monic polynomials we studied in Chapter 3 are also known as Legendre Polynomials, named after Adrien-Marie Legendre.
This is the only known "portrait" of him, which was (re)discovered in 2008.

Finished here Friday, 13 March (W09.2)

5.7 Exercises

Exercise 5.1

$\mathcal{P}_n$, the space of polynomials of degree (at most) n forms a vector space. Is it true that the space of *monic* polynomials of degree n forms a vector space?

Exercise 5.2

Given a set of $n+1$ distinct interpolation points, $\{x_0, x_1, \ldots, x_n\}$, let $\{L_i\}_{i=0}^n$ be the set of associated Lagrange basis functions.

1. Show that the set $\{L_i\}_{i=0}^n$ is linearly independent.
2. What space do they span?
3. Give two other difference bases for the same space.

Exercise 5.3

Let $\widetilde{p}_n$ be the nth Legendre Polynomial. Show that, if $n > 0$, then $\displaystyle\int_a^b \widetilde{p}_n(x)dx = 0$.

5.7 Exercises

Exercise 5.4

(i) Using the Inner Product

$$(f, g) := \int_0^1 f(x)g(x)dx,$$

find $\widetilde{p}_0(x)$, $\widetilde{p}_1(x)$, $\widetilde{p}_2(x)$ and $\widetilde{p}_3(x)$.

(ii) Find the zeros of $\widetilde{p}_2(x)$ and call them x_0 and x_1. Construct a quadrature rule for $\int_0^1 f(x)dx$ taking these as the quadrature points, and the weights as the integrals to the corresponding Lagrange polynomials.